Software Requirements Specifications

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Definition of Terms, Acronyms and Abbreviations

This section should provide the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Term	Description
ASP	Active Server Pages
RS	Requirements Specifications

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Section 1

1. Introduction

1.1 Purpose of Document

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the **Air Ticket Reservation System**.
The purpose of this document is to provide a clear, complete, and unambiguous description of the system requirements so that all stakeholders have a common understanding of the system to be developed.
This document is intended for:

- Software developers
- System designers
- Test engineers
- Project managers
- Academic advisors and evaluators

The SRS serves as a formal agreement between stakeholders and the development team and acts as a reference throughout the software development lifecycle.

1.2 Project Overview

The Air Ticket Reservation System is a **real-time, web-based application** designed to automate the process of searching, booking, and managing airline tickets.
The system provides passengers with the ability to search for flights, view real-time seat availability, make secure online payments, and receive electronic tickets without manual administrative intervention.

The system operates by integrating with **external airline reservation systems**, **payment gateways**, and **notification services** to ensure accurate, secure, and real-time processing of bookings.

Its primary goal is to improve efficiency, reduce human errors, and provide a seamless booking experience for passengers while ensuring data accuracy and system reliability.

This project reflects the architecture and behavior of modern airline reservation platforms used in the software industry.

1.3 Scope

The system will:

- Allow passengers to search for flights based on origin, destination, and travel date
- Display real-time flight schedules and seat availability
- Enable passengers to book airline tickets
- Process secure online payments through integrated payment gateways
- Generate and deliver electronic tickets via email or SMS
- Support ticket cancellation and modification according to airline policies
- Maintain booking records for passengers

The system will not:

- Manage airline staff or crew scheduling
- Handle aircraft maintenance operations
- Control airport security or boarding processes
- Manually manage flight schedules or pricing

All flight-related data, seat inventory, and booking confirmation are handled through automated external airline systems, ensuring a fully real-time operational environment.

2. Overall System Description

The Air Ticket Reservation System is a **real-time, web-based software system** designed to support automated airline ticket booking operations.
The system interacts with multiple external services such as airline reservation systems, payment gateways, and notification services to provide accurate and up-to-date flight information and booking confirmation.

This section describes the system environment, user characteristics, and constraints under which the system operates.

2.1 User characteristics

The Air Ticket Reservation System is intended for the following user categories:

Passenger (End User):

Passengers are the primary users of the system.

They are expected to have basic knowledge of web or mobile applications.

Characteristics:

- Able to use a web browser or mobile device
- No technical background required
- Can search flights, book tickets, and make payments
- Can view booking history and manage reservations

External Systems (Automated Users)

In a real-time environment, several automated systems interact with the Air Ticket Reservation System:

Airline Reservation System

- Provides real-time flight schedules and seat availability
- Confirms bookings and cancellations
- Applies airline fare and refund rules

Payment Gateway System

- Processes online payments securely
- Confirms or rejects transactions

Notification System

- Sends emails or SMS messages for booking confirmation

- Delivers electronic tickets and alerts

These systems do not require human interaction and operate automatically through secure interfaces.

2.2 Operating environment

The Air Ticket Reservation System operates in a distributed client-server environment.

Client Environment:

- Web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge
- Desktop, laptop, or mobile devices
- Internet connectivity required

Server Environment:

- Web server (Apache or IIS)
- Application server supporting RESTful APIs
- Database server (MySQL, SQL Server, or equivalent)

External Services:

- Airline reservation system APIs
- Online payment gateway services
- Email and SMS notification services

The system must be available 24/7 to support continuous booking operations.

2.3 System constraints

The Air Ticket Reservation System operates under several constraints that may limit or influence its design, implementation, and performance. These constraints are categorized as follows:

Software Constraints

- The system must integrate with third-party airline reservation systems through standardized APIs.
- Compatibility with modern web browsers is required.
- Secure communication protocols such as HTTPS must be used for all data transmission.
- The system depends on external payment gateway services for transaction processing.

Hardware Constraints

- The system requires server infrastructure capable of handling concurrent users.
- End users must have access to internet-enabled devices such as computers or smartphones.
- Performance may be affected by limitations of client-side devices.

Cultural Constraints

- The system interface must support the English language.
- Terminology used in the system should be simple and universally understandable.
- Date and time formats must be presented in internationally accepted standards.

- **Legal Constraints**
- The system must comply with aviation regulations and airline policies.
- Passenger personal and payment data must be protected according to data privacy laws.
- Secure handling of financial transactions is mandatory.

Environmental Constraints

- The system is designed for use in an online environment and requires stable internet connectivity.
- The system must operate efficiently without relying on sound-based notifications.
- The application must function properly in different physical environments such as offices or public places.

User Constraints

- The system is intended for general users with basic computer literacy.
- No technical training should be required to use the system.
- The user interface must be intuitive and easy to navigate.

Off-the-Shelf Component Constraints

- The system relies on third-party airline APIs, payment gateways, and notification services.
- Any limitations or service outages of these external components may affect system availability.
- Changes in third-party service policies or APIs may require system updates.

3. External Interface Requirements

This section describes the external interfaces required by the Air Ticket Reservation System to communicate effectively with hardware devices, software systems, and communication networks.
The system operates in a real-time environment and depends on multiple external components to provide accurate flight information, secure payment processing, and timely notifications.

At a high level, the system interacts with passengers, airline reservation systems, payment gateways, and notification services through secure and standardized interfaces.

3.1 Hardware Interfaces

The Air Ticket Reservation System is a web-based application and does not require specialized hardware.

It interfaces with standard computing devices used by end users and servers used for system hosting.

Supported hardware interfaces include:

- Desktop and laptop computers
- Mobile devices such as smartphones and tablets
- Server machines hosting the application and database

Hardware interaction characteristics:

- User input is received through keyboards, touchscreens, and pointing devices
- Output is displayed through screens in the form of web pages and electronic tickets
- Optional interaction with printers for printing booking confirmations

All hardware interactions are handled through standard operating system and browser mechanisms.

3.2 Software Interfaces

The Air Ticket Reservation System interfaces with several external software components to support real-time functionality.

Database Management System

- Stores passenger details, booking records, and transaction logs
- Ensures data consistency and reliability

Airline Reservation Systems

- Provides real-time flight schedules and seat availability
- Confirms ticket bookings and cancellations
- Exchanges data through secure APIs

Payment Gateway Systems

- Processes online payments
- Returns transaction status to the system
- Ensures secure financial communication

Notification Services

- Sends booking confirmations and electronic tickets
- Supports email and SMS delivery

Operating System and Web Server

- Runs on standard operating systems such as Windows or Linux
- Hosted on web servers such as Apache or IIS

Data exchanged includes:

- Flight details
- Passenger information
- Payment transaction status
- Booking confirmations

All software components communicate using structured data formats such as JSON or XML.

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3.3 Communications Interfaces

The Air Ticket Reservation System relies on network-based communication to interact with users and external systems.

Communication mechanisms include:

- Web browsers using HTTP/HTTPS protocols
- RESTful APIs for system-to-system communication
- Email communication using SMTP protocols
- SMS gateways for mobile notifications

Security and data transfer requirements:

- All communication must use encrypted HTTPS connections
- Secure authentication mechanisms must be implemented for API access
- Sensitive data must be transmitted in encrypted form
- Real-time synchronization is required to maintain accurate seat availability

Message formatting:

- Data exchanged between systems must follow standardized formats such as JSON
- Consistent request and response structures must be maintained

4. Functional Requirements

FR-1 Flight Search

The system shall allow passengers to search available flights based on:

- Origin airport
- Destination airport
- Travel date
- Number of passengers
- Travel class (Economy, Premium Economy, Business, First Class)

FR-2 Real-Time Flight Data Retrieval

The system shall retrieve real-time flight schedules, seat availability, and base fares from external airline reservation systems.

FR-3 Passenger Registration

The system shall allow passengers to register by providing:

- Full name
- Contact information
- Government-issued identification (if required by airline policy)

FR-4 Passenger Authentication

The system shall authenticate passengers using valid credentials before allowing booking or payment actions.

FR-5 Flight Selection

The system shall allow passengers to select a flight from the available search results based on fare, schedule, and travel class.

FR-6 Seat Availability Validation

The system shall verify real-time seat availability before allowing seat selection.

FR-7 Seat Selection Rules

The system shall allow passengers to select seats according to the following rules:

- Seats must be available at the time of selection
- Seat type availability depends on the selected travel class
- Restricted or emergency exit seats may require passenger eligibility

FR-8 Fare Calculation

The system shall calculate the total ticket fare based on:

- Selected travel class
- Base fare provided by the airline
- Applicable taxes
- Airport fees
- Fuel surcharge
- Service charges

FR-9 Dynamic Fare Rules

The system shall apply dynamic pricing rules, including:

- Fare variation based on demand
- Fare differences by booking time
- Airline-specific pricing policies

FR-10 Multi-Passenger Booking Rules

The system shall support booking for multiple passengers in a single transaction while ensuring seat availability for all passengers.

FR-11 Seat Locking Mechanism

The system shall temporarily lock selected seats during the payment process to prevent double booking.

FR-12 Booking Creation

The system shall create a booking record only after successful seat locking and fare validation.

FR-13 Payment Processing

The system shall process payments through an integrated payment gateway and support:

- Credit cards
- Debit cards
- Digital wallets

FR-14 Payment Validation Rules

The system shall confirm booking only if:

- Payment is authorized successfully
- Payment amount matches the calculated fare

FR-15 Booking Confirmation

The system shall confirm the booking after receiving confirmation from both:

- Payment gateway
- Airline reservation system

FR-16 Electronic Ticket Generation

The system shall generate an electronic ticket containing:

- Passenger details
- Flight details
- Seat number
- Fare breakdown
- Booking reference number

FR-17 Ticket Delivery

The system shall deliver the electronic ticket via:

- Email
- SMS notification (where applicable)

FR-18 Booking History Management

The system shall allow passengers to view past and upcoming bookings.

FR-19 Ticket Cancellation Rules

The system shall allow ticket cancellation subject to:

- Airline cancellation policies
- Fare class rules
- Time remaining before departure

FR-20 Refund Processing

The system shall initiate refunds based on:

- Cancellation eligibility
- Airline refund policies

- Payment gateway refund procedures

FR-21 Schedule Change Handling

The system shall notify passengers of flight schedule changes or cancellations initiated by the airline.

FR-22 Error Handling and Validation

The system shall handle and display appropriate messages for:

- Seat unavailability
- Payment failure
- Network or system errors

FR-23 Compliance Enforcement

The system shall enforce airline and regulatory rules related to passenger data and booking limitations.

5. Non-functional Requirements

NFR1 Performance

- System pages shall respond within **3 seconds**.
- Search results shall load within **5 seconds** under peak load.
- The system shall support **500 concurrent users**.
- Booking and cancellation operations shall complete within **2 seconds**.

NFR2 Maintainability

The system shall be designed to be **easily maintainable and updatable**, allowing modifications to flight data, schedules, or pricing without major downtime.

NFR3 Security

3.1 The system shall ensure **secure transmission and storage** of user data and payment information.

3.2 The system shall restrict unauthorized access and require **user authentication** before accessing personal or financial data.

NFR4 Reliability / Aberrant data

The system shall handle **invalid or unexpected inputs** gracefully by displaying appropriate error messages instead of crashing.
The system shall continue to operate correctly even if minor faults occur in some components.

NFR5 Scalability

The system shall efficiently handle **peak loads and increased traffic** during special events, holidays, or promotional periods.

NFR6 Availability

- The system shall be available **99.5% of the time**, excluding scheduled maintenance.
- Scheduled maintenance periods shall be **notified to users in advance**.

NFR7 Backup and Recovery

In the event of a system failure, data shall be **restored from backups** without loss of important information.

6. Assumptions and Dependencies

This section identifies assumptions made during the requirements analysis and external dependencies that the **Air Ticket Reservation System** relies on.
These are **not system constraints**, but factors that could impact the system if they change, are incorrect, or are not met.

6.1 Assumptions

1. **User Assumptions**
    - Passengers have access to internet-enabled devices such as computers, laptops, tablets, or smartphones.
    - Users have basic digital literacy and can navigate web or mobile applications.
    - Users will provide accurate personal and payment information when registering or booking tickets.
    - Users will adhere to airline rules and policies when booking, modifying, or canceling tickets.

2. **System Operation Assumptions**
    - External airline reservation systems provide accurate, up-to-date flight schedules and seat availability.
    - Payment gateway services are reliable and available to process transactions in real-time.
    - Notification systems (email and SMS) function correctly to deliver confirmations and alerts.
    - Network infrastructure is stable, providing sufficient bandwidth for system communication.
3. **Business Rules Assumptions**
    - Airline fare calculation policies, taxes, and service charges remain consistent with the information retrieved from the airline system.
    - Cancellation, refund, and rescheduling policies are correctly implemented by external airline systems.
    - System load will remain within expected usage patterns; extreme spikes may affect response times.
4. **Maintenance and Updates Assumptions**
    - Scheduled maintenance will be planned during low-traffic periods.
    - System updates will be deployed without affecting critical operations.
    - Security patches and software upgrades will be applied regularly to maintain system integrity.

6.2 Dependencies

1. **External Airline Systems**
    - The system depends on airline reservation systems to provide accurate flight schedules, seat availability, and fare information.
    - Any downtime or incorrect data from these systems may directly impact booking accuracy.
2. **Payment Gateway Services**
    - Integration with payment gateways is essential for processing online payments.

- The system is dependent on these services for transaction authorization and settlement.

3. **Notification Services**
   - Email and SMS services are required for delivering electronic tickets and booking notifications.
   - Any failures in these services may prevent passengers from receiving timely booking confirmations.
4. **Network and Internet Infrastructure**
   - The system relies on a stable internet connection for real-time operations.
   - Network outages or slow connectivity can impact system responsiveness.
5. **Regulatory and Legal Compliance**
   - The system depends on adherence to aviation regulations, data privacy laws, and financial transaction standards.
   - Any changes in laws or regulations may require system updates or modifications.
6. **Third-Party APIs**
   - Changes in external APIs (airline systems, payment gateways, notification services) may require system modifications.
   - Timely updates and communication from these providers are essential to maintain system functionality.

7. References

This section should provide a complete list of all documents referenced at specific point in time. Each document should be identified by title, report number (if applicable), date, and publishing organization. Specify the sources from which the references can be obtained. (This section is like the bibliography in a published book).

Ref. No.	Document Title	Date of Release/ Publication	Document Source
PGBH01-2003-Proposal	Project Proposal	Oct 20, 2003	<Give the path of your Project repository/Folder>

Ref. No.	Document Title	Date of Release/ Publication	Document Source

8. Appendices

This section should include supporting detail that would be too distracting to include in the main body of the document.